Stock Valuation

Chapter 6

Introduction to Financial Management (Pope)

Characteristics of Stock

Major financing vehicle for corporations

- Provides holders with an opportunity to share in the future cash flows of the issuer.
- Holders have “ownership” in the company (residual claim on profit)
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Stock Terminology

Common stock

- First sold (**initial public offering or IPO**) in the **primary markets**
- Most investors trade stock in the **secondary market**
- Stocks may or may not issue a **dividend** (managerial decision)
- Frequently carry a **voting right** most notably regarding Board of Directors

Reading Stock Market Data

Bid Price: the prices in which investors are willing to buy shares

Ask Price: the price at which current shareholders are willing to sell their shares

NYSE - Delayed Quote • USD

FedEx Corporation (FDX)

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267.45 +3.09 (+1.17%) **267.99** +0.54 (+0.20%)

At close: October 14 at 4:00 PM EDT | Pre-Market: 7:08 AM EDT

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1D 5D 3M 6M YTD 1Y 5Y All

5.72%

Key Events | Mountain | Advanced Chart

Previous Close	264.36	Day's Range	264.00 - 267.90	Market Cap (intraday)	65.344B	Earnings Date	Dec 19, 2024
Open	264.34	52 Week Range	224.69 - 313.84	Beta (5Y Monthly)	1.18	Forward Dividend & Yield	5.52 (2.06%)
Bid	267.55 x 1000	Volume	920,534	PE Ratio (TTM)	16.53	Ex-Dividend Date	Sep 9, 2024
Ask	268.20 x 1000	Avg. Volume	1,702,184	EPS (TTM)	16.18	1y Target Est	306.67

Stock Valuation

There are three methods discussed in the text for finding the value of common equity:

1. Valuation by comparable (ratios and multiples)
2. Free Cash Flow models (FCF)
3. Discounted dividend models (DDM)

A Simple DDM

Assume you plan to buy a stock, hold it for one year, and then sell it at the end of that year

$$PV = \frac{Div_1 + P_1}{1+r}$$

PV = present (intrinsic or fair) value of the stock
r = the discount rate
Div_1 = expected dividend at end of year
P_1 = predicted stock price at end of year

Example 1:

What is the intrinsic value of a stock if its expected dividend $4 and you anticipated to sell the stock for $62 in one year? Assume that investments of equal risk earn a return of 10%.

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$$PV = \frac{Div_1 + P_1}{1+r} = \frac{4+62}{1+0.1} = \frac{66}{1.1} = 60$$

A Simple DDM continued

If we rearrange the previous formula for r, we get an equation for expected return,

$$r = \frac{Div_1 + P_1 - P_0}{P_0} = \underbrace{\frac{Div_1}{P_0}}_{\text{Dividend yield}} + \underbrace{\frac{P_1 - P_0}{P_0}}_{\text{Capital gain yield}}$$

Discounted Dividend Models

Stock price equals the present value of all expected future dividend payments

$$P_0 = \frac{Div_1}{(1+r)^1} + \frac{Div_2}{(1+r)^2} + \cdots \frac{Div_t + P_t}{(1+r)^t}$$

What does this say about value and the holding period? What if we hold the stock for a very long time?

Constant Growth (Gordon)

If we assume 1) we will hold the stock forever and 2) the firm's dividend grows at a constant rate (g) then,

$$P_0 = \frac{Div_1}{(1+r)^1} + \frac{Div_1(1+g)}{(1+r)^2} + \cdots + \frac{Div_1(1+g)^{t-1}}{(1+r)^t}$$

$$P_0 = \frac{Div_1}{r-g}$$

Example 2:

Let's say that the Peak Growth Company just paid its shareholders an annual dividend of $2.00 and has announced that the dividends would grow at an annual rate of 8% forever. If investors expect to earn an annual rate of return of 12% on this investment how much would they offer to buy the stock for?

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$$P_0 = \frac{Div_1}{r-g} = \frac{Div_0(1+g)}{r-g}$$

$$P_0 = \frac{2\ (1+0.08)}{0.12-0.08} = 54$$

Non-constant Growth

When dividends have uneven growth, the future dividends and price must be forecasted to determine today's value

- Issues again arise when trying to forecast dividend and future prices
- Use a compassion value like P/E ration for a comparable firm or industry

$$P_0 = \frac{Div_1}{(1+r)^1} + \frac{Div_2}{(1+r)^2} + \cdots \frac{Div_t + P_t}{(1+r)^t}$$

Stock Terminology

Preferred stock

- No ownership or voting right
- Fixed dividend payment (either amount or percentage)
- No maturity (like common stock)

Preferred Stock

Preferred stock becomes a perpetuity,

$$P_0 = \frac{Div_1}{r}$$

We can also use this if we consider the firm a "no-growth" firm, where 1) the firm re-invests no profit, and 2) the firm pays out all profit as dividend (pay out EPS)

Deriving g

A **sustainable growth rate** is achieved by a firm re-investing its earnings while keeping long-term debt ratios constant

- Earnings can either issued as dividends (payout) or kept as retained earnings (plowed back)
- If earnings are retained, they would theoretically return ROE

$$g = \frac{plowed\ back\ earnings}{total\ earnings} \times \frac{total\ earnings}{initial\ equity}$$

$$g = plowback\ ratio \times ROE$$

Example 3:

Aqua America has an ROE of 12.5% and is expected to pay a dividend of $0.86. It intends to plowback 38% of its earnings and the opportunity cost of capital is 7.0%. What is the stock price?

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$$g = plowback\ ratio \times ROE = 0.38 \times 0.125 = 0.0475$$

$$P_0 = \frac{Div_1}{r-g} = \frac{0.86}{0.07-0.0475} = 38.23$$

Deriving g (continued)

A **sustainable growth rate** assumes that *ROE* and the plowback ratio are constant. If not,

- Value is only added when *ROE* is greater than *r*
- **PV of Growth Opportunity (PVGO) = Market Price – Market Value of Assets in Already in Place**

Example 4:

Recall from before, Aqua America has an ROE of 12.5% and is expected to pay a dividend of $0.86. It intends to plowback 38% of its earnings and the opportunity cost of capital is 7.0%. Its price should be $38.23. If Aqua America decides to payout all future earnings as dividends, what will happen to its stock price?

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$$EPS = \frac{Div_1}{payout\ ratio} = \frac{Div_1}{(1 - plowback\ ratio)} = \frac{0.85}{(1 - 0.38)} = 1.37$$

$$P_0 = \frac{EPS}{r} = \frac{1.37}{0.07} = 19.57$$

$$PVGO = 38.23 - 19.57 = 18.66$$